

AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous ML discovery through **pure evidence**.

PERSISTENT WORLD

FRESH SCIENTIST

EVIDENCE FIRST

Problem → System → Experience → Evaluation → Insights

Can an agent carry a research task to completion?

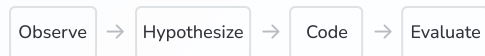
Turn a task contract into a tested model, a reproducible implementation, and evidence another researcher can inspect.

Explore	Choose hypotheses, write code, and run experiments.	Scientific judgment
Recover	Handle failures and revise unsuccessful ideas.	Long-running work
Deliver	Retain a valid model and the evidence behind it.	Verifiable progress

Our test case: short-video ranking on KuaiRand-Pure, with a fixed evaluator and public-validation boundary.

Why research agents fall short

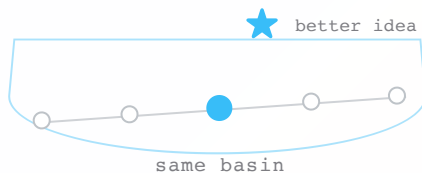
01 WORKFLOW



Fixed recovery paths

Unexpected errors need human help.

02 SINGLE AGENT



History shapes search

Errors can persist

03 TREE SEARCH

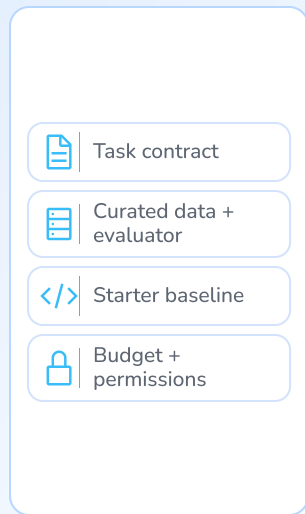


Repeated context reads

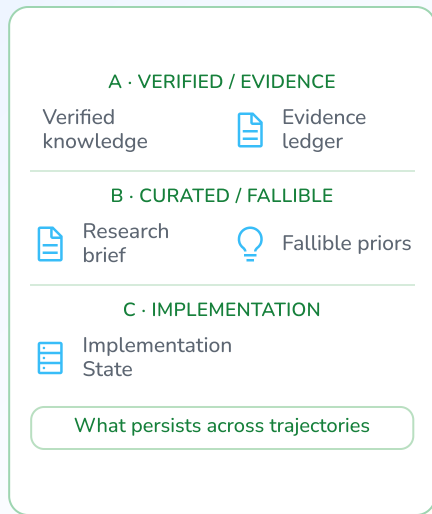
Each small edit pays the reading cost.

A system for sustained research

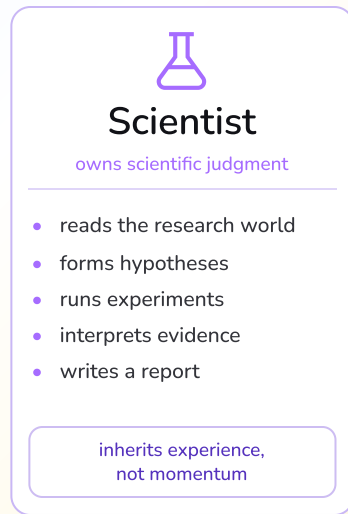
Environment



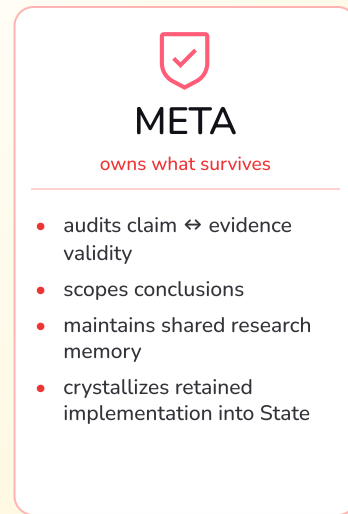
Research World



Fresh Scientist



META Review



Selective persistence / trajectory handoff

audit → scope → preserve

Coding-agent harness + runtime



Workspace isolation



Runner + evaluator



Logging



Artifact capture



State operations

SciOdyssey

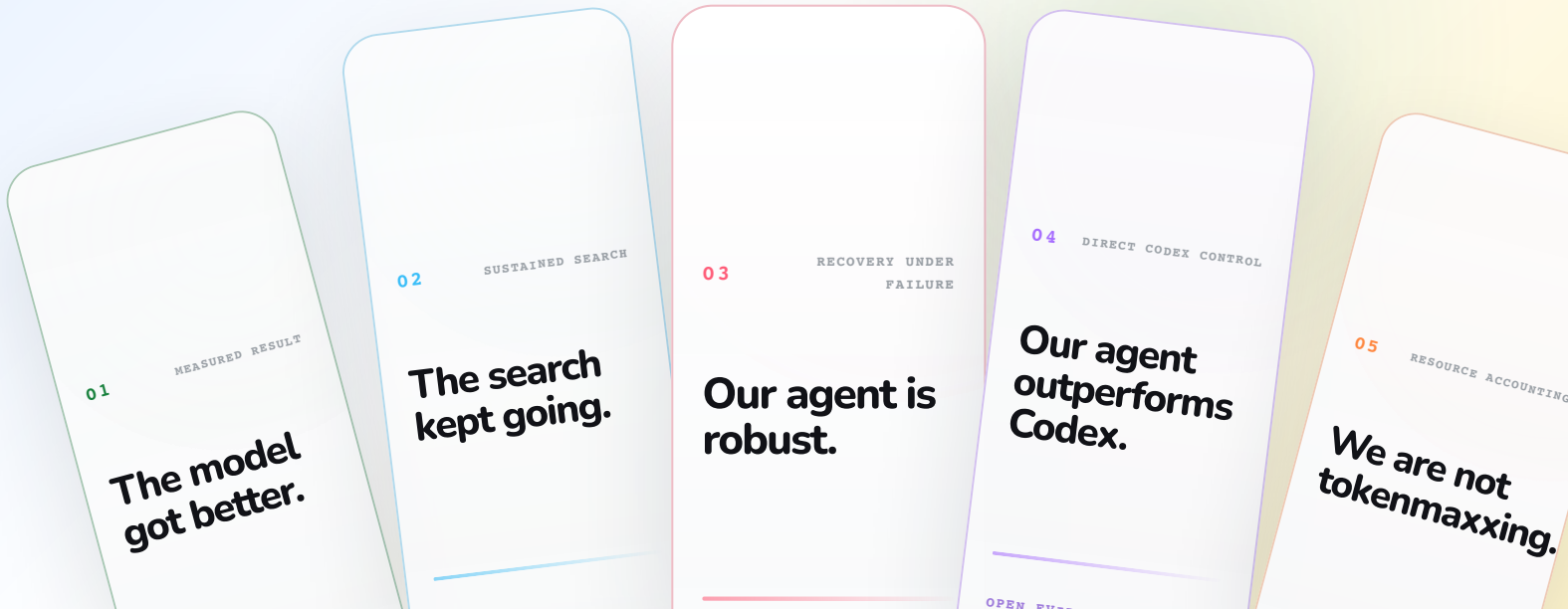
A research layer over your agent harness.

Your agent. One persistent research world.

CLAIM

Our agent can carry a research task to completion.

KUAIRAND-PURE • FIXED EVALUATOR • PUBLIC VALIDATION



The run taught us more than the final score.

One durable lesson from the experiments.

01 Diversity is a prior.

Keep `gemini-3.7-flash` as META while Scientist rotates across `gpt-5.6-sol`, `gemini-3.7-flash`, and `gpt-5.6-luna`. This widens priors; it does not prove the gain.

A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

01 MANY AGENTS



Give every agent a branch.

main

clean trunk

● agent A / branch

● agent B / branch

● agent C / branch

review

→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

02 NO QUOTA / NO LAPTOP



Leave the next move in GitHub.

01

Issue /
prompt

inject the task



02

GitHub
Action

run the agent



03

Commit

review the change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

Keep the experience. Reopen the search.

```
# From the configured repo, run one autonomous cycle
$ ./scripts/research-agent step --cli codex \
  --target /absolute/path/to/project --allow-edits
```

github.com/zc6600/research-agent-kuairand ↗

Code · Technical report · Experiment ledger · Checked output

Research is an **open-world** task.
We need an **open-world** coding agent.

01 OPEN ACTION SPACE



Shell



Files



Browser



Git

Use what exists

02 SELF-RECOVERY



Read errors



Inspect docs



Inspect
source



Install
packages

Fix what breaks

03 RUNTIME EXTENSION



MCP



Skills



CLI



Packages

Add what's missing

WORKFLOW EXAMPLE • LANGGRAPH

LangGraph makes a known workflow explicit as nodes + edges.

Missing capability? Leave the graph → inspect → extend → retry

Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD



Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture ·
integration



Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles ·
evidence review



Shilin Xu

DATA / METRICS

Data workflow · contracts
· alignment



GE GAO

RUNTIME / RELIABILITY

Execution · testing ·
integration support



Jiran Li

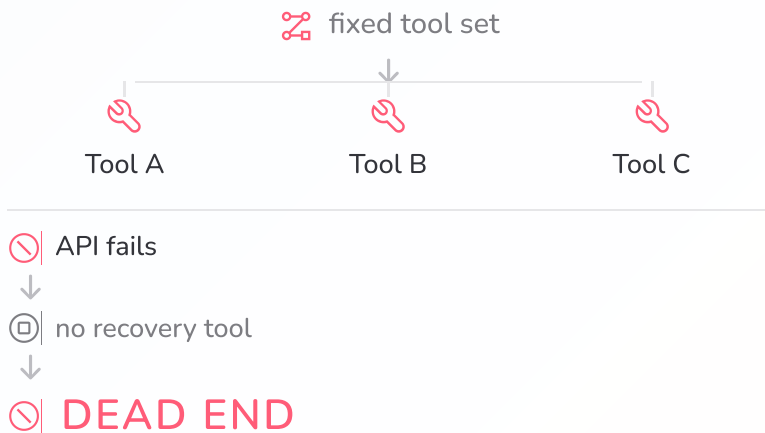
EVIDENCE /
REPRODUCIBILITY

Telemetry · documentation
· packaging

Research is an **open-world** task.
We need an **open-world** coding agent.

LANGGRAPH-STYLE ORCHESTRATION

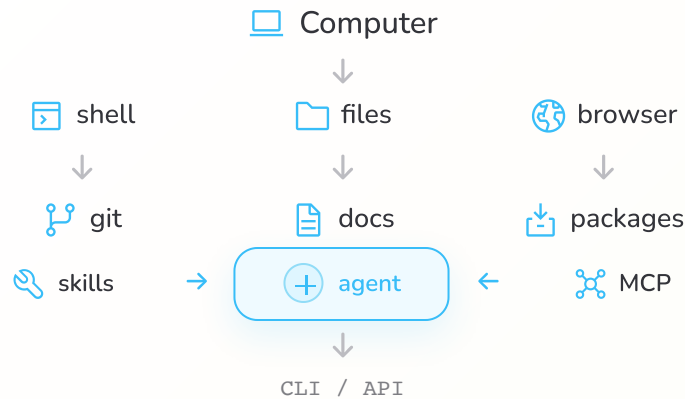
predefined graph



Capabilities are bounded by what developers anticipated.

CODING-AGENT HARNESS

open action space



fail → inspect → learn → modify environment → retry

The agent can acquire the missing capability at runtime.